mass number 100 and another of mass number 130, for the sum total of the binding energies of these two nuclei is greater than the binding energy of the original single nucleus of mass number 230. Instances of such splitting —*fission*—of atoms were actually observed by Hahn and Strassmann in 1938.

As a matter of fact, we might well wonder why it is that not every one of the heavier elements is an alpha emitter or splits into two nuclei of more or less equal size, instead of holding together, as they do according to actual experience, for a remarkably long time at least. As regards fission, the lifetimes of these elements are even longer. We shall inquire into this question in our sixth lecture.